Spatio-Temporal Crime Occurrence Prediction and Crime Type Forecasting in Chicago Using Interpretable Machine Learning Models

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***Abstract*—Urban crime forecasting is a challenging spatio-temporal prediction problem because reported crime patterns vary across location, time of day, and recent historical activity. This study uses reported crime incidents from the Chicago Crime Dataset between January 1, 2024 and December 31, 2025 to develop two prediction tasks: crime occurrence prediction and grouped crime type forecasting. The original event-only dataset was transformed into a supervised Community Area by Hour dataset, producing approximately 1.35 million hourly observations with both crime and no-crime examples. Temporal, cyclical, lag-based, and rolling-window features were engineered to capture short-term crime patterns. Logistic Regression, Random Forest, Histogram Gradient Boosting, and a Neural Network MLP were evaluated using a chronological non-shuffled train-test split. For crime occurrence prediction, Histogram Gradient Boosting achieved the strongest threshold-tuned F1-score of 0.525, while Random Forest achieved the highest recall of 0.629. For grouped crime type forecasting, Random Forest achieved the best balanced performance with a Macro-F1 score of 0.400. The results show that occurrence prediction is more reliable than grouped crime type forecasting using only spatial and temporal features. The paper also discusses limitations related to reported-crime bias, missing external context, and responsible use of predictive models in public-safety settings.**

***Index Terms*—crime prediction, spatio-temporal forecasting, machine learning, Chicago crime data, public safety analytics, interpretable machine learning**

I. Introduction

Urban crime analysis is an important application area for data science because crime patterns often vary across location, time of day, day of week, and neighborhood context. Public crime datasets provide an opportunity to study these patterns using machine learning methods. However, crime forecasting is difficult because reported crime incidents are sparse, imbalanced, and affected by social, spatial, and reporting factors.

This study focuses on short-term crime forecasting in Chicago using public crime records from the Chicago Data Portal. The work is organized around two prediction tasks. The first task is crime occurrence prediction, where the goal is to predict whether at least one crime will occur in a given Chicago community area during a given hour. The second task is grouped crime type forecasting, where the goal is to predict the broader category of crime once a crime occurrence exists.

A key challenge is that the original dataset contains only reported crime events. It does not directly contain no-crime examples. To address this, the event-level data is converted into a complete Community Area by Hour grid. This creates a supervised learning dataset containing both crime and no-crime observations. The resulting dataset allows binary crime occurrence prediction while preserving spatial and temporal structure.

The goal of this study is to examine how far a transparent and reproducible machine learning pipeline can go using only publicly available reported-crime data and engineered spatio-temporal features. Rather than proposing a new deep learning architecture, this work focuses on the full modeling process: converting event-only incident records into a supervised forecasting dataset, constructing temporal and historical features, comparing multiple model families, and evaluating the results using a chronological non-shuffled split.

The main contributions are:

- A Community Area by Hour supervised dataset construction from event-only Chicago crime records.
- Temporal, cyclical, lag-based, and rolling-window feature engineering for short-term crime forecasting.
- A comparison of Logistic Regression, Random Forest, Histogram Gradient Boosting, and Neural Network MLP models.
- Separate evaluation of crime occurrence prediction and grouped crime type forecasting.
- Discussion of interpretability, limitations, and ethical concerns related to predictive crime modeling.

II. Related Work

Crime prediction has commonly been studied as a spatio-temporal forecasting problem, where historical incidents are aggregated over spatial units and time intervals to estimate future risk [1]. Prior research has used statistical models, traditional machine learning, deep learning, and graph-based methods to capture spatial dependence and temporal trends.

Recent studies have applied more complex neural architectures to urban crime forecasting. For example, transformer-style temporal models and spatial graph convolutional networks have been used to model community-level crime patterns in Chicago [2]. Other recent work has explored graph convolutional networks for crime hotspot prediction using spatial relationships between regions [3]. These methods are useful when richer spatial graph structure and larger modeling complexity are desired.

In contrast, this study focuses on a simpler and more reproducible empirical pipeline. The objective is not to outperform advanced graph neural networks, but to evaluate how interpretable feature engineering and standard model families perform under a realistic chronological split. This makes the results easier to reproduce and easier to explain in an applied setting.

Interpretability is an important consideration in public-safety modeling because prediction outputs may influence sensitive decisions. Prior work has used interpretable machine learning methods, including feature importance and SHAP-style explanations, to better understand model behavior in spatial and urban prediction tasks [4], [5]. This study follows the same motivation by analyzing the role of temporal, lag-based, rolling-window, and community-area features.

Crime prediction also raises ethical concerns. Reported-crime datasets do not represent all crime; they represent incidents that were reported, recorded, and processed. Historical crime data may reflect reporting differences, enforcement patterns, and neighborhood-level inequalities. Prior research and policy discussions have warned that predictive policing systems can reinforce existing biases if deployed without careful oversight [6], [7]. For this reason, the present study treats crime forecasting as an aggregate academic analysis task and does not propose individual-level or operational policing use. This positioning is important because the present work is intended as a reproducible applied machine learning study rather than a replacement for more complex graph-based or deployment-oriented public safety systems.

III. Dataset and Preprocessing

The dataset used in this study is the Chicago Crime Dataset from the Chicago Data Portal [8]. Records were filtered from January 1, 2024 to December 31, 2025. The raw dataset contained approximately 494,273 crime incident records and 22 attributes, including date, location, community area, district, primary crime type, arrest status, domestic flag, latitude, and longitude.

Records with missing geographic coordinates were removed, leaving approximately 492,776 usable records. The date field was converted into a timestamp, and temporal attributes such as hour, day of week, month, day, and weekend indicator were extracted.

A major preprocessing challenge is that the original dataset contains only reported crime events. In its raw form, each row represents a crime that already occurred. Therefore, the data cannot directly be used for binary crime occurrence prediction because it does not include examples where no crime occurred.

To address this, the event-level dataset was transformed into a complete Community Area by Hour grid. Each row in the new dataset represents one Chicago community area during one hourly time slot. If one or more crimes occurred in that community area during that hour, the binary target was set to 1. Otherwise, the target was set to 0. This transformation converted the original incident-level dataset into a supervised spatio-temporal forecasting dataset with both crime and no-crime observations.

The final crime occurrence dataset contained 1,352,736 hourly community-area observations. This structure allowed the models to learn from both positive and negative examples while preserving spatial and temporal ordering.

For the second task, crime type forecasting was performed only on rows where a crime occurred. Original Chicago primary crime types were grouped into three broader categories: Property, Violent, and Other. The grouped category label was created before model training so that all models were evaluated on the same multiclass target. This grouping reduced label sparsity and made the multiclass task more stable. Rows with no crime occurrence were excluded from crime type forecasting because a crime category is undefined when no crime occurred.

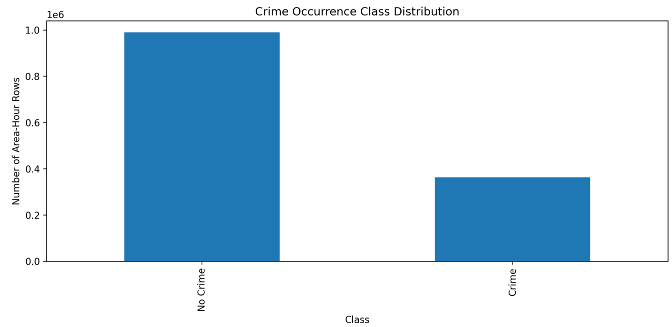

Fig. 1. Distribution of crime and no-crime observations after Community Area by Hour dataset construction.

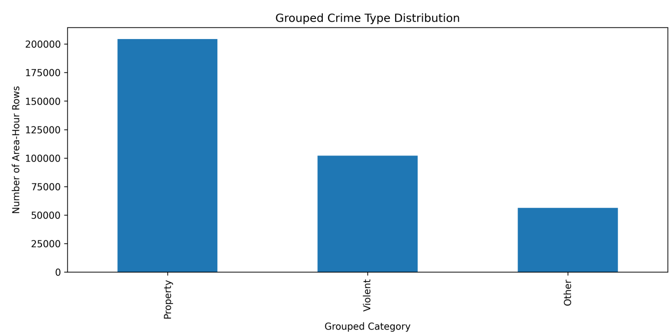

Fig. 2. Distribution of grouped crime type categories used for multiclass forecasting.

IV. Feature Engineering

The feature engineering process was designed to represent spatial location, calendar time, cyclical time patterns, and recent historical crime activity. These features were created separately for the crime occurrence dataset and then reused for grouped crime type forecasting where applicable.

Spatial information was represented using the Chicago community area identifier. Temporal information included hour of day, day of week, month, day of month, and a weekend indicator. These variables help capture common temporal patterns such as differences between weekdays and weekends or changes in activity across different hours of the day.

Because hour of day is cyclical, the raw hour value was transformed using sine and cosine encoding. This prevents the model from treating hour 23 and hour 0 as far apart, even though they are adjacent in real time. The cyclical hour features were computed as:

$$hour\_sin = \sin\left(\frac{2\pi \times hour}{24}\right) \tag{1}$$

$$hour\_cos = \cos\left(\frac{2\pi \times hour}{24}\right) \tag{2}$$

To capture short-term historical crime patterns, lag features were created within each community area. These features represented whether crime occurred in the same community area during previous time windows. The lag features included one-hour, two-hour, three-hour, and twenty-four-hour lag values.

Rolling-window features were also created to summarize recent crime activity over multiple time horizons. These included rolling mean values over three-hour, six-hour, twelve-hour, and twenty-four-hour windows. These features help the model capture local temporal momentum, such as whether recent crime activity in a community area is associated with higher near-future risk.

The final feature set included community area, calendar-based features, cyclical hour encodings, lag features, and rolling mean features. This design allowed the models to use both current temporal context and recent historical activity while keeping the pipeline simple and reproducible.

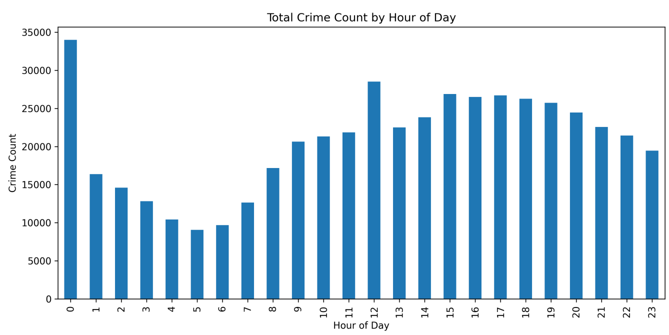

Fig. 3. Reported crime count by hour of day.

V. Methodology

The experiments were designed to compare several model families under the same preprocessing, feature engineering, and evaluation setup. A chronological non-shuffled train-test split was used for both prediction tasks. This is important for spatio-temporal forecasting because random splitting can allow future patterns to appear in the training set, leading to overly optimistic results.

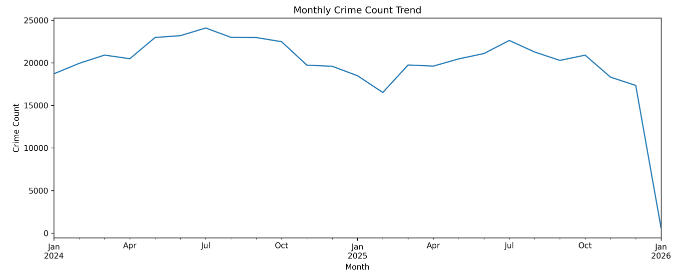

Fig. 4. Monthly reported crime trend during the study period.

For each task, the data was sorted by hourly time slot. The first 80% of time periods were used for training, and the remaining 20% were used for testing. This setup better reflects a forecasting scenario, where models are trained on earlier historical data and evaluated on later unseen time periods.

Four model families were evaluated:

- Logistic Regression
- Random Forest
- Histogram Gradient Boosting
- Neural Network MLP

Logistic Regression was used as a simple linear baseline. Random Forest was included as a non-linear tree-based ensemble method. Histogram Gradient Boosting was evaluated because gradient boosting methods often perform well on tabular data. A Neural Network MLP was included as a neural baseline to compare against traditional machine learning methods.

For crime occurrence prediction, the task was treated as binary classification. The evaluation metrics included accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC. Because the dataset is imbalanced, F1-score and PR-AUC were considered more informative than accuracy alone. Threshold tuning was also performed to identify the probability threshold that maximized F1-score on the test predictions.

For grouped crime type forecasting, the task was treated as multiclass classification. The evaluation metrics included accuracy, Macro-F1, and Weighted-F1. Macro-F1 was emphasized because it gives equal weight to each class and is useful when one crime category appears more often than others.

All experiments were implemented in Python using scikit-learn and standard data science libraries. The same chronological split was used across models to support a consistent comparison. Large raw data files and trained model artifacts were excluded from version control, while preprocessing scripts, feature engineering code, metric tables, and generated figures were organized for reproducibility.

VI. Experimental Results

This section presents the results for the two prediction tasks. For crime occurrence prediction, the main objective was to identify whether a crime would occur in a given community area and hour. For grouped crime type forecasting, the objective was to predict the broader type of crime after a crime occurrence was observed.

A. Crime Occurrence Prediction

The crime occurrence prediction results are summarized in Table I.

TABLE I
CRIME OCCURRENCE PREDICTION RESULTS

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Random Forest	0.693	0.433	0.629	0.513	0.737	0.505
Logistic Regression	0.704	0.441	0.568	0.496	0.721	0.483
Hist. Gradient Boosting	0.767	0.607	0.266	0.370	0.748	0.516
Neural Network MLP	0.764	0.607	0.231	0.335	0.733	0.499

Histogram Gradient Boosting achieved the highest accuracy, ROC-AUC, and PR-AUC among the evaluated models. It obtained an accuracy of 0.767, ROC-AUC of 0.748, and PR-AUC of 0.516. However, at the default threshold of 0.5, its recall was relatively low at 0.266. This indicates that the model was more conservative in predicting crime occurrence.

Random Forest achieved the highest recall of 0.629 and the strongest default-threshold F1-score of 0.513. This makes it useful in settings where missing potential crime-occurrence cases is more costly than producing additional false positives. Logistic Regression performed competitively as a simple baseline, while the Neural Network MLP achieved high accuracy but lower recall and F1-score at the default threshold.

Threshold tuning improved the performance of the probability-based models. Histogram Gradient Boosting achieved the best threshold-tuned F1-score of 0.525 at a threshold of 0.27. This suggests that the model's ranking ability was stronger than its default-threshold classification performance.

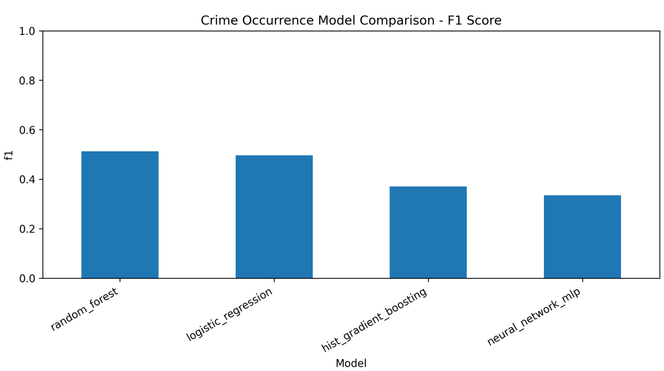

Fig. 5. F1-score comparison for crime occurrence prediction models.

B. Grouped Crime Type Forecasting

The grouped crime type forecasting results are summarized in Table II. In this task, the model predicts whether the crime belongs to the Property, Violent, or Other category.

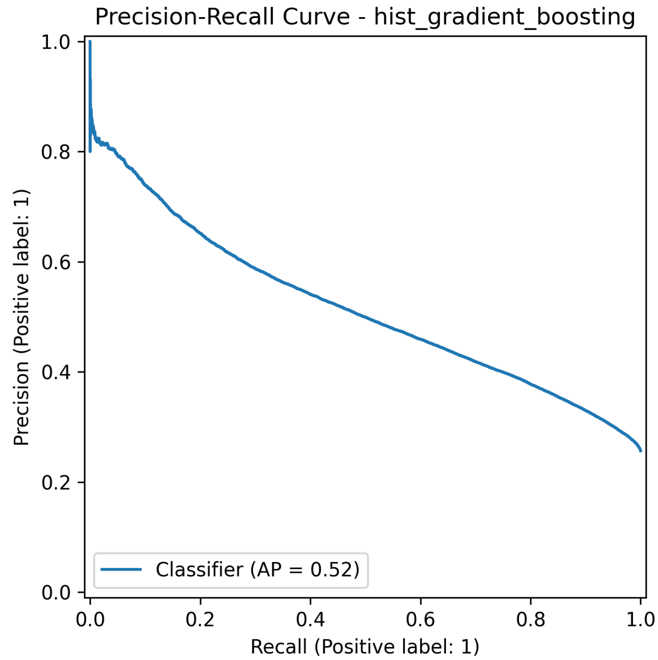

Fig. 6. Precision-recall curve for Histogram Gradient Boosting on crime occurrence prediction.

TABLE II
GROUPED CRIME TYPE FORECASTING RESULTS

Model	Accuracy	Macro-F1	Weighted-F1
Random Forest	0.439	0.400	0.458
Logistic Regression	0.438	0.369	0.446
Hist. Gradient Boosting	0.570	0.273	0.434
Neural Network MLP	0.569	0.247	0.417

Grouped crime type forecasting was more difficult than crime occurrence prediction. Histogram Gradient Boosting achieved the highest accuracy of 0.570, but its Macro-F1 score was only 0.273. This gap suggests that the model likely favored the majority crime category and performed less consistently across minority classes.

Random Forest achieved the best balanced performance, with a Macro-F1 score of 0.400 and a Weighted-F1 score of 0.458. Although its accuracy was lower than Histogram Gradient Boosting, its stronger Macro-F1 indicates better class-balanced prediction. For this reason, Random Forest was considered the most balanced model for grouped crime type forecasting.

These results show that predicting whether a crime may occur is more feasible using the current spatio-temporal feature set than predicting the specific grouped crime category. Crime type forecasting likely requires richer contextual information beyond recent crime history and calendar-time features.

VII. LIMITATIONS AND ETHICAL CONSIDERATIONS

This study has several limitations. First, the dataset contains reported crime incidents, not all crimes that occurred.

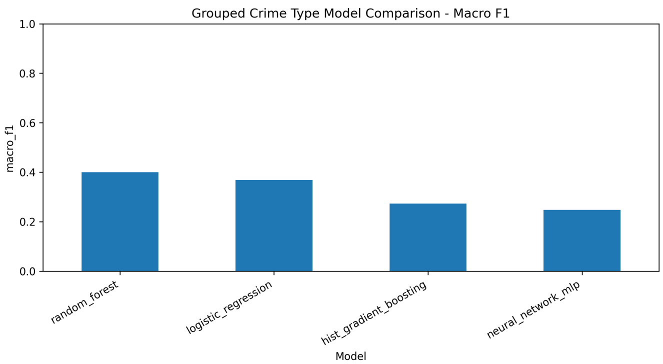

Fig. 7. Macro-F1 comparison for grouped crime type forecasting models.

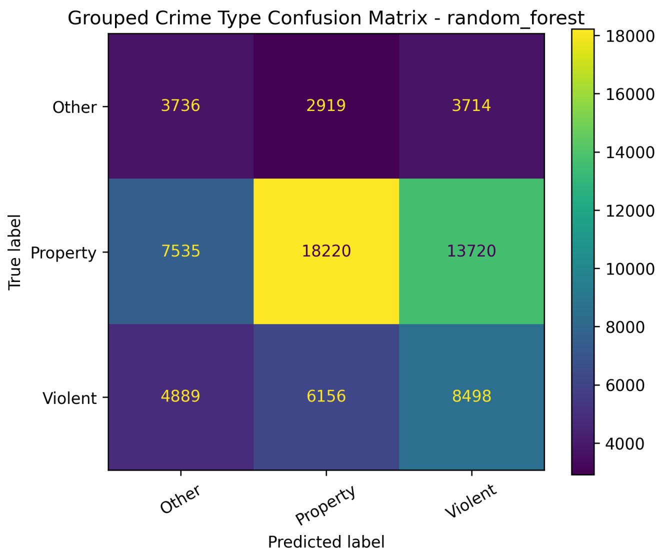

Fig. 8. Confusion matrix for Random Forest grouped crime type forecasting.

Some incidents may be underreported, while others may be more visible due to neighborhood reporting behavior, policing patterns, or public-service access. Therefore, the model learns patterns from recorded incidents rather than true underlying crime occurrence.

Second, the spatial unit used in this study is the Chicago community area. Community areas are useful for aggregation and interpretation, but they may hide smaller block-level or street-level differences. Two locations within the same community area may have different risk patterns that are not captured by the current feature set.

Third, the current feature set is limited to spatial identifiers, calendar-time variables, cyclical hour encodings, lag features, and rolling-window features. The model does not include external context such as weather, public events, school schedules, holidays, transit activity, population density, demographics, land use, points of interest, or police deployment patterns. These missing variables may explain why grouped crime type forecasting was more difficult than crime occurrence prediction.

Fourth, this study evaluates the models on one city and one recent time period. Although the chronological non-shuffled split provides a more realistic evaluation than random splitting, the results may not generalize to other cities, older time periods, or future conditions without additional validation.

Crime prediction also requires careful ethical handling. Historical crime records can reflect enforcement bias, reporting bias, and unequal policing patterns. If models trained on such data are used without safeguards, they may reproduce or amplify existing inequalities. For that reason, this work should be interpreted only as an aggregate academic analysis of reported-crime patterns.

The models developed in this study are not intended for individual-level prediction, surveillance, automated policing decisions, or direct officer deployment. Any real-world use would require fairness evaluation, bias audits, privacy protections, transparency, human oversight, and community review. The results should be used to understand broad spatio-temporal patterns in reported incidents, not to label individuals or communities as inherently criminal.

VIII. Reproducibility and Data Availability

The dataset used in this study is publicly available through the Chicago Data Portal. The raw dataset was not included directly in the repository because of file size considerations. Instead, the repository provides instructions for downloading the data and placing it in the expected local directory structure. The project repository is publicly available at: https://github.com/Dilipsingh1234/chicago-crime-spatiotemporal-prediction. The archived project release is available through Zenodo with DOI: https://doi.org/10.5281/zenodo.20384166.

The project code was organized into preprocessing, feature engineering, model training, evaluation, and figure-generation scripts. Generated result tables and selected figures were stored separately for paper preparation. The same chronological non-shuffled train-test split was used across all models to make the comparison consistent.

To support reproducibility, the repository includes the Python requirements file, model training scripts, metric outputs, and paper-ready figures. The processed datasets and trained model artifacts can be regenerated from the raw data using the provided scripts.

IX. Conclusion and Future Work

This study presented a reproducible machine learning pipeline for spatio-temporal crime occurrence prediction and grouped crime type forecasting using reported Chicago crime data. The original event-level dataset was transformed into a Community Area by Hour supervised learning dataset, allowing the models to learn from both crime and no-crime observations.

The results show that crime occurrence prediction is more reliable than grouped crime type forecasting using the current feature set. For crime occurrence prediction, Histogram Gradient Boosting achieved the strongest overall ranking and threshold-tuned performance, with a best threshold F1-score of 0.525. Random Forest achieved the highest recall and the strongest default-threshold F1-score, making it useful when identifying more potential crime-occurrence cases is preferred.

For grouped crime type forecasting, Random Forest achieved the best balanced performance based on Macro-F1 and Weighted-F1. Although Histogram Gradient Boosting achieved higher accuracy, its lower Macro-F1 score showed that accuracy alone was not sufficient for evaluating this task. This finding highlights the difficulty of predicting crime category using only spatial and temporal history.

Overall, the study suggests that engineered temporal, lag-based, and rolling-window features can support short-term crime occurrence prediction, but crime type forecasting requires richer contextual information. Future work should include external variables such as weather, public events, holidays, population density, mobility patterns, demographics, land use, and points of interest. Future research could also compare graph-based and sequence-based deep learning models under the same chronological evaluation setup.

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